

Urban Flood Monitoring and Visualization System Based on Citizen Science

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Urban flooding and the formation of flash floods represent a recurring problem in the Metropolitan Area of Asunción, affecting mobility, infrastructure, and public safety during intense rainfall events. The information available about affected areas is usually dispersed or limited to isolated reports, making real-time monitoring and analysis difficult. In this context, citizen science approaches make it possible to complement traditional systems through the active participation of the population by reporting observed events [1, 2].

The development of an integrated system for urban flood monitoring and visualization based on citizen science is proposed. The work focuses on the design and implementation of an interactive web platform oriented toward the registration and geospatial visualization of citizen reports related to water accumulation. The platform allows users to register events through web forms, incorporating descriptions, photographs, and the geographic location of incidents. For the development of the system, technologies such as Next.js and React were used for the frontend; Supabase and PostgreSQL for data management; Leaflet for interactive cartographic visualization; and Supabase Storage for image storage [3].

The results obtained during the initial implementation made it possible to validate the operation of the proposed architecture and the correct integration of the different modules. An interactive map capable of representing geolocated reports through dynamic markers and severity filters was successfully developed. Likewise, a real-time report and image storage system was implemented, enabling the centralized management of the collected information. The platform demonstrated the feasibility of integrating citizen participation with geospatial visualization tools within a web environment accessible from mobile devices and computers. In addition, the developed structure allows future extensions focused on spatial event analysis and historical information processing.

The system demonstrates the potential of citizen science applied to urban flood monitoring by integrating technological tools for the organization and visualization of user-generated geospatial information [1, 2]. As part of future stages of the work, the incorporation of automated geolocated data acquisition mechanisms from open digital sources is intended, as well as the integration of water level and rainfall sensors to complement citizen reports with real-time hydrometeorological information. Furthermore, efforts will be made to strengthen the geospatial analysis and collaborative monitoring capabilities of the system to support the identification of critical areas and contribute to urban risk management associated with flood events.

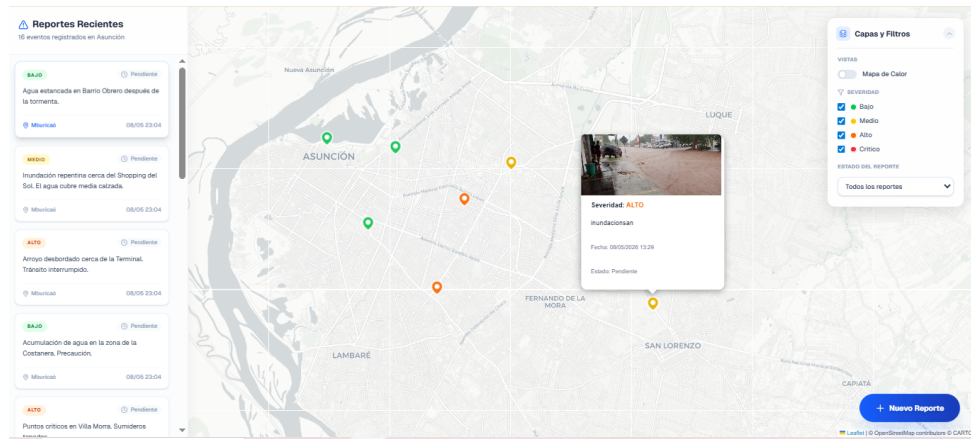


Figure 1: Interactive map developed for the visualization of citizen reports.

References

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